DS & AI Lab Project (BSDA4001)

TerraWatch: Intelligent Satellite Image Change Detection and Analysis

MILESTONE 3: MODEL SELECTION AND SYSTEM ARCHITECTURE

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BS in Data Science and Applications

GROUP 7

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1. Introduction

TerraWatch identifies visible building and road changes between two satellite images and produces descriptions linked to visual evidence. Milestone 3 establishes the model choices, module interfaces, training configuration and complete input-to-output architecture [1]. The system combines pixel-level detection, domain-specific captioning, structured claims, crop-based evidence and an accept/reject/defer referee.

The data contract uses the Milestone 2 strict LEVIR-MCI package: **9,544 pairs**, partitioned into **6,444 training, 1,242 validation and 1,858 test records**. Source masks and five reference captions support supervised learning and evaluation. Inference consumes ordered images and model-generated intermediates.

2. Overall System Architecture

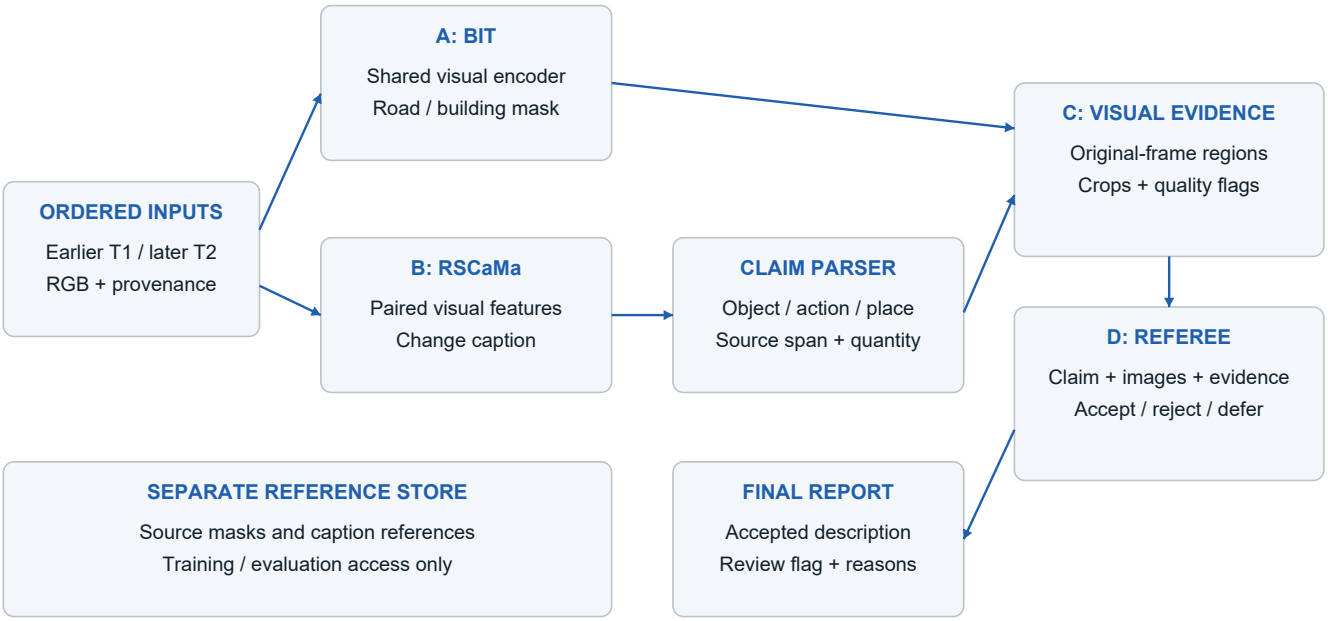


Figure 1. Detection and captioning operate on the same image pair. Predicted regions and caption claims meet at the evidence stage. Evaluation references occupy a separate store.

Layer	Technology and role
Data and persistence	Python, NumPy, Pillow; immutable image sources, split manifests, JSON/JSONL records and file hashes
Detection and description	PyTorch; BIT three-class detector and RSCaMa captioner in dedicated model environments
Evidence and decisions	OpenCV, Pydantic and isolated Transformers workers; explicit coordinates, typed records and decision gates
Execution and interaction	Colab/WSL/local CLI; two ordered image paths and a configuration file produce a JSON report and crop evidence

Model weights are acquired from their author repositories or Hugging Face. The inference design runs through local workers with cached weights. Evidence retrieval uses run IDs and file paths; the task does not require a vector database or document-retrieval stage.

3. End-to-End Workflow

Stage	Input	Output and control
Validate the pair	Earlier T1 and later T2 RGB images	Matching dimensions, temporal-order declaration and SHA-256 provenance; incompatible pairs raise an input error
Detect changes	Normalised temporal tensors	Background/road/building mask; nearest-neighbour restoration to original image dimensions
Generate and parse	Full-frame images; captioner output	Caption plus atomic object/action/location/quantity claims, source spans and parser provenance
Localise and inspect	Claim, predicted class mask and original images	Matching connected region, padded T1/T2 crops, transform record, visibility flags and structural observations
Referee and gate	Claim, fallible evidence and original images	Raw accept/reject/defer decision; probability validity and operating-threshold checks
Compose and store	Gated claim records	Accepted-description text, review flag, per-claim reasons, raw responses and auditable report JSON

Connected components identify candidate regions; they are not building-instance counts. Claim locations use the image-relative 3 x 3 grid. Matching regions retain original-T1 half-open pixel coordinates. Unlocalised claims use full-image context and receive a review reason.

The implementation executes each learned stage in a separate process, releases its memory on exit and records request, response and log files. A new run directory preserves earlier deliveries. Worker timeouts and missing referee outputs yield explicit error/review records. Insufficient visibility, an unsupported legal inference or an exact count without independent instance evidence produces a defer decision.

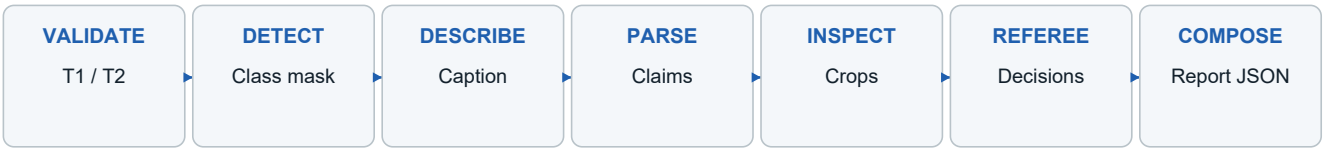


Figure 2. The operating pipeline retains intermediate evidence and raw decisions. Report composition uses accepted claim spans and carries review requirements into the final output.

4. Model Architecture Selection

Module	Selected architecture	Size, representation and integration
A - change detection	BIT with a shared ResNet18 backbone and three-class output head [2]	Four semantic tokens per temporal stream; one token-encoder layer and eight decoder layers. Class indices: 0 background, 1 road, 2 building.
B - change description	RSCaMa with CLIP ViT-B/32, spatial/temporal state-space fusion and a Transformer language decoder [3]	7 x 7 patch features, 768-dimensional fusion, three fusion layers, eight attention heads and one decoder layer; greedy output capped at 42 tokens.
C - visual evidence	Custom crop and structural witness	Parameter-free OpenCV processing: LAB luminance mean shift, Sobel edges, rectilinear orientation, local anomalies, visibility and registration diagnostics.
D - claim referee	Qwen3-VL-4B-Instruct image-backed judge; Qwen3-4B-Instruct-2507 shared-evidence comparator [4,5]	Approximately 4B language-model scale; original images plus structured evidence for VL, evidence text for the text comparator. Closed accept/reject/defer outputs.

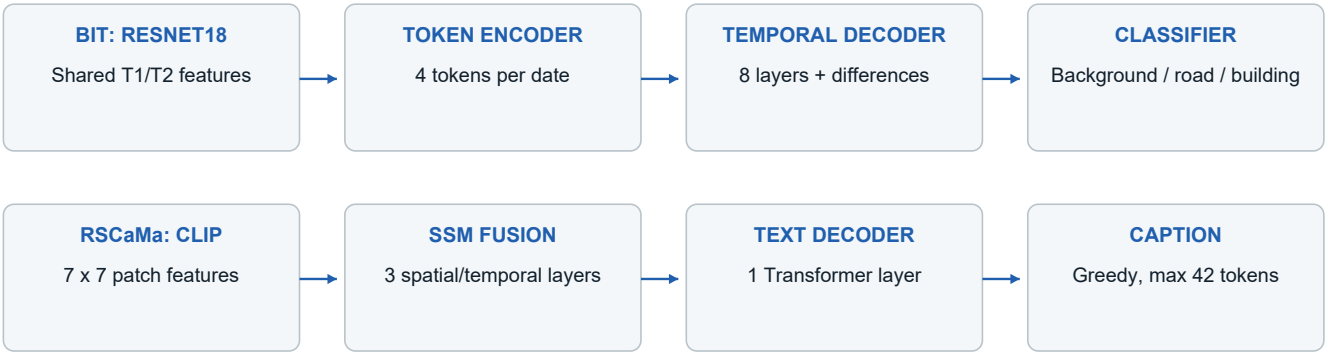


Figure 3. BIT predicts dense class targets. RSCaMa translates paired visual features into text. These modules remain independently replaceable and trainable.

BIT is the detector architecture selection, justified by its compact shared backbone and temporal context mechanism. Detector training uses ImageNet visual initialisation and a new three-class task head. RSCaMa's tested interface consumes two RGB images; detector masks enter the downstream claim-localisation and verification path. This distinction preserves the actual captioner input contract.

The selected captioning comparison includes the released RSCaMa checkpoint and a subset-trained RSCaMa implementation initialised from public CLIP weights. Both referees use the exact checkpoint revisions recorded in the supplied run metadata.

5. Justification and Experimental Evidence

The captioning comparison uses a fixed **100-pair LEVIR screen**, including **40 unchanged pairs**. The original B0 results and Sarthak Saini's local rerun use automatic claim/reference checks. Strict precision counts all extracted claims; in-scope precision evaluates claims covered by the annotation scope. Cells show **B0 / local** values [6,9].

Captioning configuration	Strict precision (%)	In-scope precision (%)	BLEU-4
Released RSCaMa	78.69 / 78.69	88.89 / 88.89	0.6223 / 0.6223
RSCaMa, 20%-budget retrain	79.17 / 76.04	87.96 / 82.02	0.5786 / 0.5793
Qwen3-VL-4B	58.33 / 55.70	72.92 / 71.74	0.0267 / 0.0183
SmolVLM2-2.2B	38.58 / 37.76	48.41 / 47.13	0.0102 / 0.0000
Pixel-difference template	69.00 / 69.00	69.00 / 69.00	0.0000 / 0.0000
Always no change	40.00 / 40.00	40.00 / 40.00	0.2798 / 0.2798

RSCaMa is the selected captioner. Its released checkpoint reproduces the stored B0 aggregates at the reported precision and leads the local screen on both precision measures and BLEU-4. Local ROUGE-L is **0.7464**, CIDEr-D **1.3184** and reference-fact recall **76.8%**, with **0/40 unchanged-pair false alarms**. The retrained model records 5/40 false alarms, Qwen3-VL 11/40 and SmolVLM2 33/40. The no-change control also records zero alarms, but its 40% strict precision demonstrates the need to measure change coverage alongside false positives.

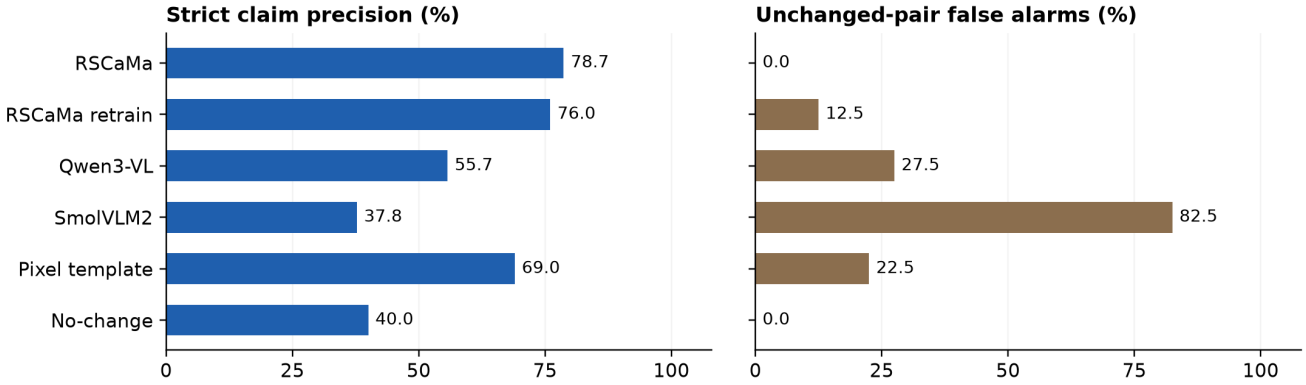


Figure 4. Automatic claim precision and false alarms on 40 unchanged pairs. Higher precision and lower false-alarm rates are preferred.

The reproducibility screen retains the raw-release selection and upstream vocabulary construction from official-split captions. The strict 9,544-pair Milestone 2 package supplies the separate training-data contract. The controlled retrain uses **1,313 training pairs and 50 development pairs**, with a training-subset vocabulary. The original Apple M4/MPS batch-one screen records median caption times of **0.139 s** for RSCaMa and **7.605 s** for Qwen3-VL; these timings retain their original hardware context.

5.1 Detector, Transfer and Referee Evidence

The local **BIT-ResNet18 two-epoch experiment** uses **100 training pairs, 25 validation pairs and a separate 100-pair evaluation population**, including 45 unchanged pairs. Both detector outputs are reduced to binary any-change predictions for the F1 comparison [9].

Detector	Binary any-change F1	Unchanged-pair false alarms
BIT-ResNet18	45.8%	31/45 (68.9%)
Otsu pixel-difference control	16.7%	45/45 (100.0%)

BIT improves binary F1 by **29.1 percentage points** and reduces false alarms by 14 unchanged pairs. Its road/building class-mean IoU is **26.8%**; the Otsu control's binary change IoU is **9.1%**. These IoUs describe different target spaces and are reported separately. BIT's compact backbone, temporal context and three-class output support region localisation. The measured false-alarm rate motivates downstream visual checking.

Released RSCaMa transfer reruns use 100 pairs per source:

Transfer source	Strict / in-scope precision	BLEU-4
SECOND-CC	48.04% / 50.00%	0.2426
WHU-CDC	55.56% / 66.67%	0.4717

These aggregates match the stored transfer results at the displayed precision. Source dependence supports separate reporting and evidence-grounded verification. WHU usage follows its recorded source-rights disposition.

The L4 referee screen uses **30 validation pairs and 150 constructed annotation-proxy claims**, evaluated before the 0.8 acceptance gate.

Referee configuration	Proxy claims	Raw decision agreement
Qwen3 text, common evidence	150	41.3%
Qwen3-VL, common evidence	150	45.3%
Qwen3-VL, image-backed	150	56.7%

Image-backed Qwen3-VL achieves the strongest agreement in this controlled comparison and is the selected visual referee. The text judge supplies a shared-evidence comparator. Caption checks, detector overlap and referee agreement measure distinct component behaviours; each retains its own evaluation population.

6. Model Inputs and Outputs

Contract	Representation
Detector input/output	Paired B x 2 x 3 x 256 x 256 tensors; BIT scaling to [-1,1]; B x 3 x 256 x 256 logits and categorical mask
Captioner features	Author CLIP processor: 224-pixel RGB views, CLIP mean/std; temporal patch features and checkpoint-matched word vocabulary
Structured claim	Object, action, optional exact quantity, image-relative location and exact caption source span
Evidence	Original-frame box, padded crops, transform metadata, mask/witness producer IDs, quality flags and fallible observations
Referee/report	Checkpoint chat template, closed decisions, raw response, probability semantics, gated outcome, reason codes and accepted-description text

The RSCaMa retrain builds its vocabulary from training-subset captions and preserves start/end/padding tokens. The released checkpoint uses its matching vocabulary. Qwen uses the publisher tokenizer/processor and temporal-order prompt. Reference captions, human labels and source masks are excluded from inference requests.

7. Training Strategy

Component	Training configuration and checkpoint rule
BIT task training	ImageNet visual initialisation; backbone frozen for five epochs, then trained at 1e-5; temporal/head parameters at 1e-4. AdamW, weight decay 0.01; cosine schedule, 50-epoch cap; batch eight with four-step accumulation.
Detector loss/selection	Equal mixture of class-weighted focal loss (gamma 2; background/road/building weights 0.10/0.45/0.45) and soft Dice. Validation macro-IoU selects checkpoints; patience eight. Paired detector flips preserve masks.
RSCaMa retraining	Public CLIP visual initialisation; temporal fusion and caption decoder trained through the upstream trainability policy. Token cross-entropy, Adam 1e-4, batch 12 pairs, 40-epoch cap, patience eight and seed zero. Development BLEU-4 selects best weights.
Witness and referee	Structural witness uses fixed image-processing rules. Qwen weights remain frozen; prompt, image budget, model revision and the 0.8 operating gate are recorded.

The detector row specifies the task-training configuration. The two-epoch experiment in Section 5.1 uses its own 100/25/100-pair protocol; the 50-epoch configuration defines the full task-training profile. The captioner row records the implemented subset-retraining script. Training and validation retain their fixed source partitions; test records are reserved for evaluation. Checkpoint files and histories preserve model configuration, epoch and selection score.

Colab supplies GPU training/inference, with L4-class evidence from the returned decision sessions. Model workers use separate environments because captioner and referee library versions differ. Sequential residency and stored intermediates control memory without loading the complete model stack simultaneously.

8. Pipeline Implementation and Verification

The pipeline entrypoint accepts two image paths and a model configuration. Separate workers execute BIT, RSCaMa and the referee; the coordinator performs claim extraction, predicted-region matching, crop inspection and final composition. Model checkpoints, upstream source directories and worker interpreters are explicit configuration fields. This keeps the architecture reproducible across Colab, WSL and local environments.

Acceptance requires a valid decision record and the configured operating gate. Missing evidence and invalid probability vectors produce review reasons. Exact counts require independent instance evidence; connected-region counts do not satisfy that requirement. Output descriptions concern visible changes and preserve uncertainty through the defer state.

Verification evidence	Executed scope
Local component suite	**83 passing tests**, including the Torch-dependent shim checks, in the supplied local run summary
Visual witness	50/50 synthetic outcomes; 2/3 coarse mask-region-presence checks across three authored real-image cases over two source pairs
Captioning	Six configurations rerun on 100 LEVIR screening pairs; two 100-pair transfer reruns
Decision exports	Six checksum-verified ZIPs, five session IDs and 40 distinct image pairs; successful Qwen text/VL outputs and preserved failure records
Orchestration	CPU test doubles exercise localisation, reference isolation, decision gating, failure review and preservation of existing run directories

The real-image witness checks measure coarse mask-region presence. Claim wording, structural observations and referee decisions retain separate fields. Decision-session exports use a frozen Qwen visual witness. Component protocols retain their run identities and separate metrics.

Human assessment uses SUPPORTED, CONTRADICTED, UNSUPPORTED and UNVERIFIABLE judgements from two independent raters. Agreement and Cohen's kappa precede third-rater adjudication using the original images and contextual crops. SUPPORTED maps to accept, CONTRADICTED/UNSUPPORTED to reject and UNVERIFIABLE to defer. This measures claim factuality separately from mask overlap, reference-caption similarity and automated proxy decisions.

The architecture fixes the complete data flow, model choices and training controls. The delivered scripts provide model-worker interfaces, detector training, evidence records and auditable output generation, supported by the consolidated experiment register.

References

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3. Liu, C. et al. (2024). *RSCaMa: Remote Sensing Image Change Captioning With State Space Model*. IEEE GRSL. https://doi.org/10.1109/LGRS.2024.3404604 ; https://github.com/Chen-Yang-Liu/RSCaMa

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Team Sign-off

The team has reviewed and approved this Milestone 3 report.

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Priyanshu Agarwal	PA	10 November 2026
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